

SUMMARY TABLE: BIOLOGY GRADE 11

Sub-Strand 3.1: Reproduction in Animals — Teacher Reference (Pre-filled)

SUMMARY TABLE: BIOLOGY GRADE 11	
Sub-Strand	Sub-Strand 3.1: Reproduction in Animals
Driving Question	DRIVING QUESTION: How do different animals ensure successful fertilisation and development of offspring, and how can science assist when human reproduction fails?

INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 Frogs, Hens and Cows: Many Ways to One Goal – How Do Animals Reproduce?	Learners compared videos, images and cards of frogs spawning in a pond near Kisumu, a hen (kienyeji chicken) laying and brooding eggs, and a cow giving birth on a Kericho dairy farm. Teacher note: Ask learners to record where fertilisation happens (outside or inside the body), how many eggs or offspring are produced, and how much parental care is given. Allow WAIT TIME of 5 seconds after each question, then cold-call 3 groups to share their tables. Common prediction to expect: 'All animals lay eggs.' Use the evidence to challenge this.	Fertilisation can be external (frogs and fish release gametes into water) or internal (hens, cows, humans, where sperm are deposited inside the female). External fertilisation needs water so that gametes do not dry out and sperm can swim to the eggs. Because many gametes are lost, animals with external fertilisation produce very large numbers of eggs. Internal fertilisation protects gametes and embryos, so fewer offspring are produced and survival rates are higher. Development can be oviparous (eggs hatch outside, as in hens), viviparous (young develop inside the mother, as in cows) or ovoviparous (eggs hatch inside the mother). Teacher note: Stress that fewer offspring with more care is a strategy, not a 'better' or 'worse' one.	The anchoring phenomenon is that different animals reproduce in different ways yet all succeed. Model answer: A frog lays hundreds of eggs in water because predators, drying and failed fertilisation destroy many, so a large number raises the chance that some survive. A cow produces one calf at a time because sperm meet the egg inside the body, the embryo is protected and nourished, and the calf is cared for after birth. Both strategies ensure the species survives. Teacher note: Accept answers that link the environment (water vs land) to the number of offspring and level of protection. Add the new question 'How do the body parts make internal fertilisation work?' to the Driving Question Board.
Lesson 2 Built for Reproduction: Dissecting Mammalian Reproductive Structures	Learners dissected (or studied labelled models, charts and photographs of) the reproductive systems of a male and a female mammal, such as a rat or a goat organ set from a local butchery, following safety and ethics rules (gloves, hand washing, respect for specimens). They	Each part of the mammalian reproductive system is adapted to its function. The testes produce sperm and testosterone, and the ovaries produce ova and the hormones oestrogen and progesterone. The sperm ducts and urethra transport sperm, while the accessory glands add fluids that nourish	Link to the phenomenon: successful reproduction in a cow or in humans depends on structures that make gametes, bring them together and protect the embryo. Model answer: Sperm made in the testes travel through the epididymis and sperm duct, are deposited in the vagina, and swim

	identified testes, epididymis, sperm duct (vas deferens), urethra, ovaries, oviducts, uterus, cervix and vagina, and noted the shape, size and texture of each. Teacher note: Prepare alternatives (diagrams, models) where specimens are not available. Cold-call 4 learners to point out and name a structure and say what they think it does, with WAIT TIME of 5 seconds.	and help sperm swim. The oviduct (fallopian tube) is lined with cilia that sweep the ovum along and is the usual site of fertilisation. The uterus has a thick, muscular wall and a rich lining that supports implantation and growth of the embryo. The cervix is a narrow muscular ring that keeps the foetus in during pregnancy and the vagina receives sperm and serves as the birth canal. Teacher note: Reinforce the principle 'structure fits function' with every organ.	up through the cervix and uterus to meet the ovum in the oviduct, where cilia and muscle movement carry the ovum toward them. The fertilised egg moves to the uterus, which has a thick lining for implantation. If the oviduct is blocked, fertilisation or implantation fails. Teacher note: This sets up later discussion of IVF. Add learner questions on the Driving Question Board, e.g. 'What does a sperm cell look like and how does it reach the egg?'
Lesson 3 Built for the Journey: Structure of Sperm and Ovum	Learners examined prepared slides, micrographs or diagrams of sperm and ovum, then drew and labelled them. They also watched a short animation or read a flow chart comparing natural fertilisation in the oviduct with in vitro fertilisation (IVF) in a laboratory dish. Teacher note: Ask learners to note each structure on the sperm (head, acrosome, nucleus, mid-piece with mitochondria, tail) and on the ovum (nucleus, cytoplasm with food reserves, cell membrane, jelly coat). Use 5 seconds of WAIT TIME, then cold-call 3 learners to link one structure to one job.	The sperm is adapted for movement and fusion: the tail (flagellum) propels it, the mid-piece contains many mitochondria that release energy, the acrosome holds enzymes that digest the ovum's protective layers, and the haploid nucleus carries the father's genes. The ovum is large, with cytoplasm rich in food reserves, a jelly coat that protects it and helps prevent entry of more than one sperm, and a haploid nucleus carrying the mother's genes. During fertilisation the sperm nucleus fuses with the ovum nucleus to form a diploid zygote. In IVF, ova and sperm are combined in a laboratory dish and the resulting embryo is placed in the uterus, whereas in natural fertilisation this happens in the oviduct. Teacher note: Watch for the misconception that many sperm fuse with the ovum.	Link to the phenomenon: fertilisation succeeds only when both gametes are structurally suited to their roles. Model answer: Millions of sperm are released but only a few reach the ovum; those with energy from mitochondria and working tails get there, and the acrosome enzymes let one sperm penetrate the jelly coat. The ovum's food reserves feed the zygote during early development. IVF can help when the gametes cannot meet naturally, for example because of blocked oviducts, since the dish takes the place of the oviduct. Teacher note: Add questions to the Driving Question Board such as 'What controls when the ovum is released?' and update the class model with gamete structures.
Lesson 4 Hormonal Conductors: How Pituitary and Ovarian Hormones Regulate the Menstrual Cycle and Egg Release	Learners analysed graphs of FSH, LH, oestrogen and progesterone levels over a 28-day cycle, together with diagrams of follicle development, ovulation, corpus luteum formation and the thickness of the uterine lining. Teacher note: Guide learners to match peaks on the graph to events (e.g. the LH surge just before ovulation around day 14). Use a think-pair-share with 10 seconds of WAIT TIME, then cold-call 4 pairs to explain one pattern they noticed.	The pituitary gland (FSH and LH) and the ovaries (oestrogen and progesterone) interact through negative and positive feedback to time follicle development, ovulation and preparation of the uterus. FSH stimulates follicle growth and oestrogen secretion. Rising oestrogen repairs and thickens the uterine lining, and at high levels it triggers (positive feedback) an LH surge that causes ovulation. After ovulation the follicle becomes the corpus luteum, which secretes progesterone to maintain the lining and inhibit FSH and LH (negative feedback). If no pregnancy occurs the corpus luteum breaks down, hormone	Link to the phenomenon: successful reproduction needs precise timing so that an ovum is available when sperm are present and the uterus is ready. Model answer: FSH from the pituitary grows a follicle, the follicle releases oestrogen, oestrogen peaks and triggers the LH surge, ovulation occurs, and progesterone then maintains the uterine lining for implantation. Hormonal imbalance, for example too little FSH or LH, can prevent ovulation and cause infertility, which helps explain why some couples need medical help. Teacher note: Encourage respectful, scientific discussion of menstrual health, and add

		levels drop and menstruation follows. Teacher note: Clarify that day 14 is an average and cycles vary between individuals.	'How do hormones keep a pregnancy going?' to the Driving Question Board.
Lesson 5 From Fertilisation to Birth: How Hormones and the Placenta Sustain Pregnancy	Learners studied a timeline and diagrams of fertilisation, cleavage, implantation, placenta formation, foetal growth and birth, and examined a diagram of the placenta showing the umbilical cord, the maternal and foetal blood vessels and the exchange surface. They also read data on hormone levels during pregnancy. Teacher note: Ask learners to trace what moves from mother to foetus and what moves back. Use 5 seconds of WAIT TIME and cold-call 3 learners for each direction of exchange.	Hormones (hCG, progesterone, oestrogen, oxytocin, prolactin) coordinate pregnancy, birth and lactation. hCG from the early embryo keeps the corpus luteum secreting progesterone, which maintains the uterine lining and prevents menstruation. Later the placenta produces progesterone and oestrogen. Oxytocin stimulates uterine contractions during birth and milk ejection during breastfeeding, and prolactin stimulates milk production. The placenta exchanges oxygen, nutrients and antibodies from mother to foetus, and carbon dioxide and urea from foetus to mother, by diffusion, without the two blood supplies mixing. Teacher note: Mention that harmful substances such as alcohol, nicotine and some viruses (e.g. HIV) can cross the placenta, linking to health and responsible behaviour.	Link to the phenomenon: after fertilisation, development of offspring inside the mother depends on hormones and the placenta. Model answer: The embryo implants in the thick uterine lining and releases hCG, so progesterone stays high and the pregnancy continues. The placenta's large, thin, well-supplied surface lets oxygen and nutrients such as glucose reach the foetus and removes wastes. At birth oxytocin strengthens contractions through positive feedback, and after birth prolactin and oxytocin support breastfeeding, which nourishes the baby. Teacher note: Update the class model to include the placenta and hormones, and add questions about what happens when conception or pregnancy does not occur naturally.
Lesson 6 Helping Where Nature Needs a Hand: Debating IVF, Surrogacy and Sperm Banking	Learners read short case studies of Kenyan couples facing infertility (for example, blocked oviducts, low sperm count, or a missing or damaged uterus) and matched each case to a biotechnology. They then took roles in a structured class debate on IVF, surrogacy and sperm banking, considering scientific, ethical, cultural, religious and cost issues. Teacher note: Set ground rules for respectful discussion and be sensitive to learners with family experience of infertility. Give 10 seconds of WAIT TIME for preparation, then cold-call 4 learners to present one evidence-based argument each.	IVF, surrogacy and sperm banking are biotechnologies that address specific structural or hormonal causes of infertility. In IVF, fertilisation happens in a laboratory dish: ova are collected, often after hormone treatment to stimulate several follicles, fertilised with sperm, and the embryo is transferred to the uterus. It helps with blocked oviducts or low sperm counts. In surrogacy, another woman carries the pregnancy when the intended mother cannot. Sperm banking stores sperm by freezing for later use, for example before cancer treatment. Each technique has benefits and concerns, including cost, access in Kenya, emotional stress, legal and ethical issues, and cultural beliefs. Teacher note: Emphasise that infertility has many causes in both men and women and is not the fault of one person.	Link to the phenomenon: when natural fertilisation or development fails, science can assist by replacing or supporting the specific step that has failed. Model answer: A woman with blocked oviducts can have her ova fertilised in a dish and the embryo placed in her uterus, since her ovaries and uterus work. A woman without a functional uterus may need a surrogate. A man starting chemotherapy can store sperm beforehand. Learners should justify their positions with biological evidence and also acknowledge ethical and cultural views. Teacher note: Accept different opinions if they are supported by reasons, and add remaining questions to the Driving Question Board.
Lesson 7	Learners reviewed the Driving Question	What we learned: the types of fertilisation	Final answer to the Driving Question. Model

Shared Decisions, Hormonal Health and Biotechnology: Our Final Explanation of How Animals Reproduce	Board, their model drafts from earlier lessons and their notes, then applied their ideas to a new scenario or case (for example, a farmer in the Rift Valley improving breeding in dairy cattle, or a couple weighing options for infertility). They revised their class model and prepared a final written or oral explanation. Teacher note: Use a gallery walk so that groups compare models; allow 10 seconds of WAIT TIME and cold-call 4 groups to justify one change they made to their model.	and development (external, oviparous, viviparous, ovoviparous); how structure relates to function in reproductive organs and gametes; how FSH, LH, oestrogen and progesterone regulate the menstrual cycle and ovulation; how hCG, progesterone, oestrogen, oxytocin and prolactin together with the placenta sustain pregnancy, birth and lactation; and how IVF, surrogacy and sperm banking can assist when reproduction fails, along with the ethical issues they raise. Teacher note: Check that learners can connect these ideas rather than list them separately.	answer: Animals ensure successful fertilisation and development by strategies suited to their environment: frogs and fish release many gametes in water, while hens and mammals use internal fertilisation with protected eggs or embryos and parental care. In mammals, specialised organs and gametes, precise hormonal timing of ovulation, and the placenta and pregnancy hormones allow fertilisation and development to succeed. When human reproduction fails because of a structural or hormonal problem, science can assist with IVF, surrogacy or sperm banking, but decisions must also weigh health, cost, culture, religion and ethics. Teacher note: Assess using a rubric covering evidence, accurate use of terms, links between structure and function, and respectful consideration of different viewpoints.
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ATTRIBUTION AND LICENSING

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